SVM, KNN & Full Week Comparison

Day 33 | PM Session — Week 6, Machine Learning & AI

IIT Gandhinagar | PG Diploma in AI-ML & Agentic AI Engineering

Part A: 8-Algorithm Comparison (CV Accuracy, Test Accuracy, Training Time, Feature Importances)
Part B: TF-IDF + LinearSVC vs Logistic Regression (Text Classification, Top Words)
Part C: Interview — High-Dim Analysis (Q1) & SVM Overfitting Diagnosis (Q3)
Part D: Algorithm Selection Guide (Verified + Edge Cases)

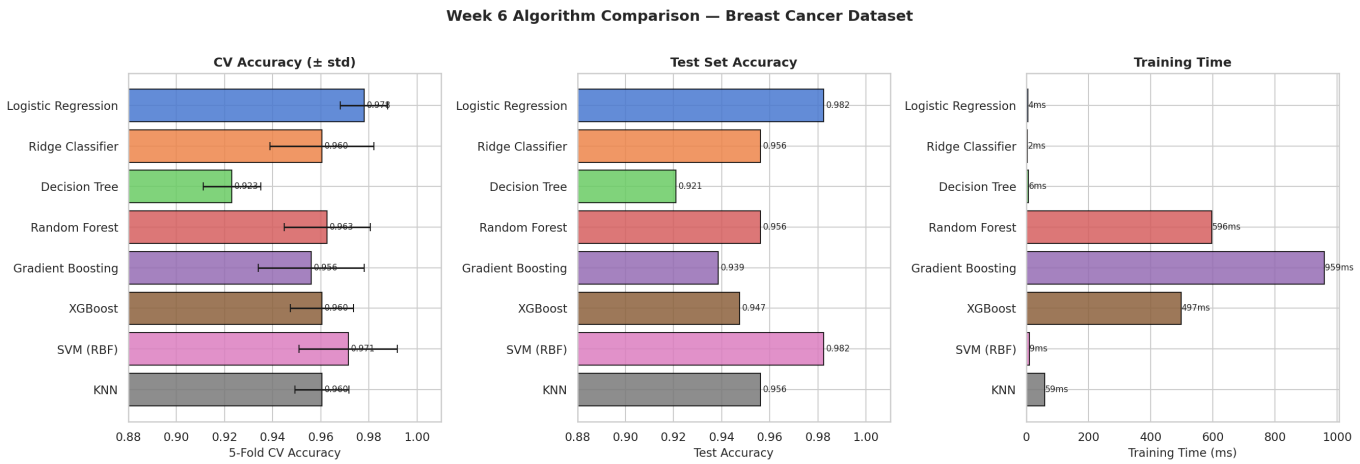

Figure 1: 8-Algorithm Comparison — CV Accuracy, Test Accuracy & Training Time (Part A)

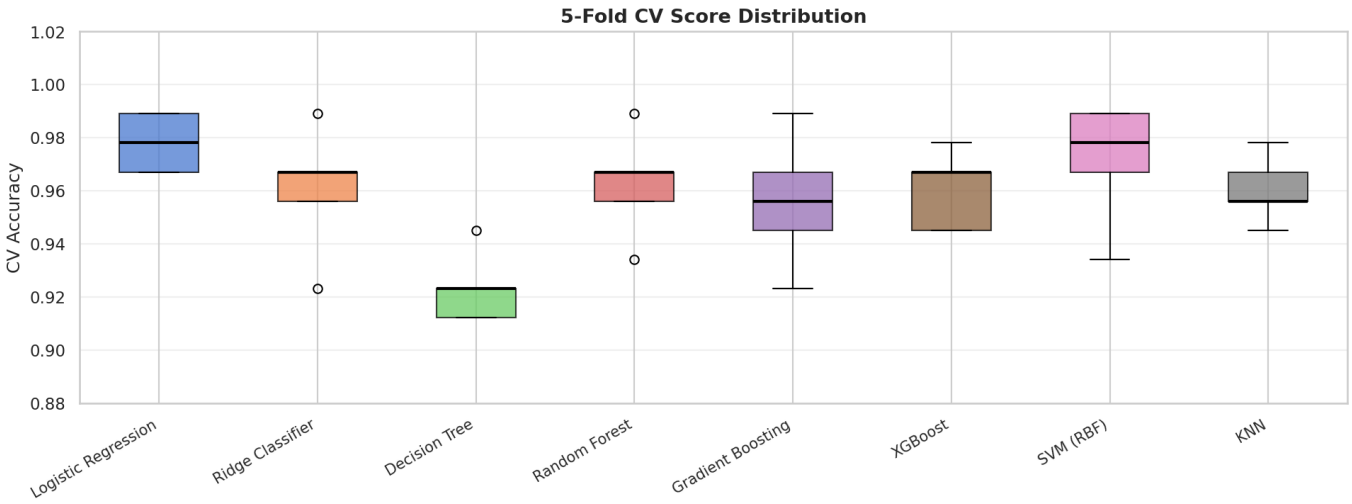

Figure 2: 5-Fold CV Score Distributions — Stability Analysis (Part A)

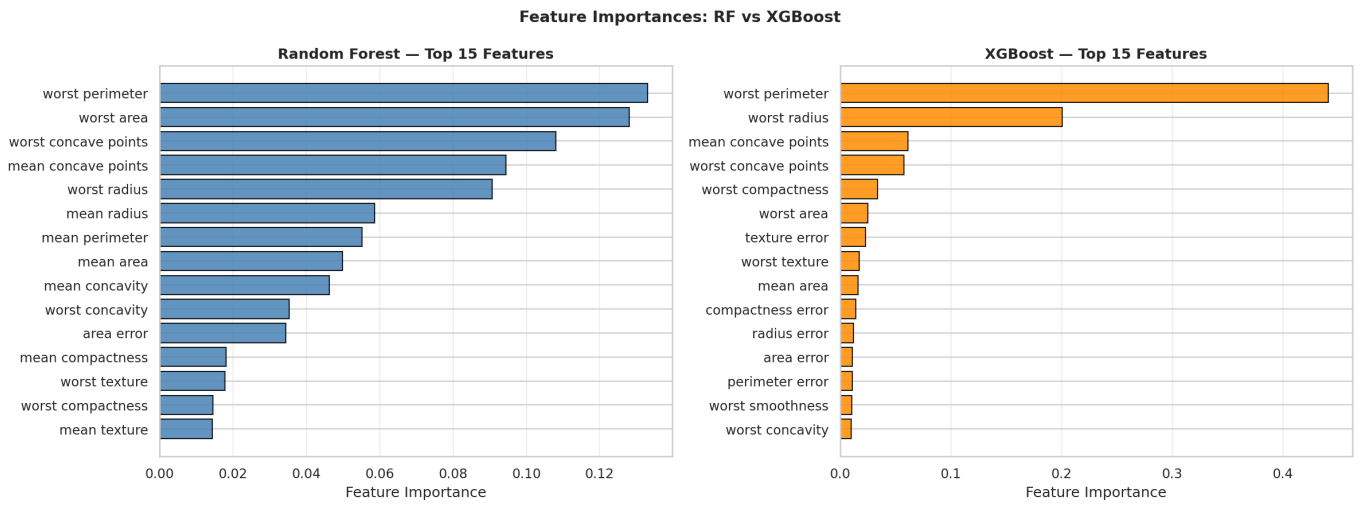

Figure 3: Feature Importances — Random Forest vs XGBoost Top 15 (Part A)

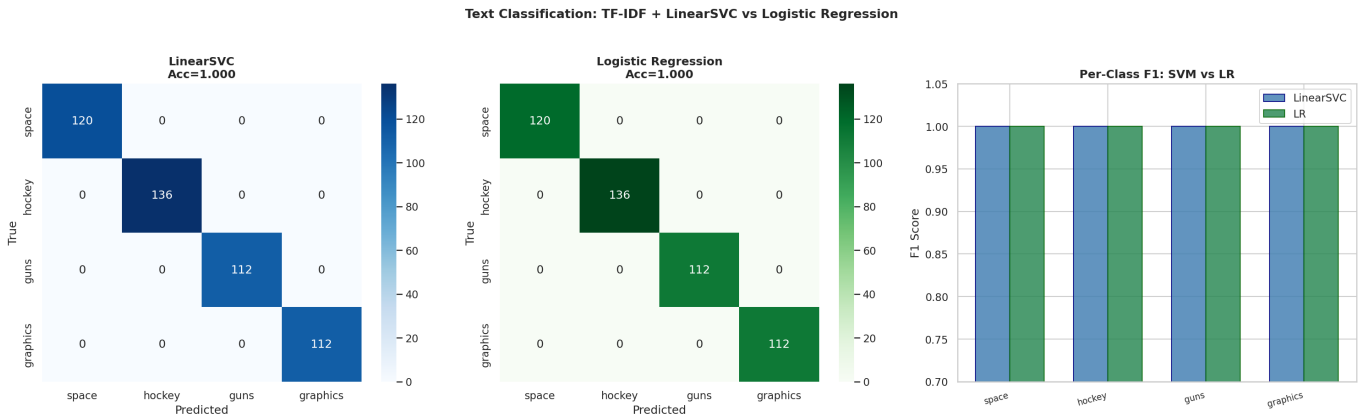

Figure 4: Text Classification — LinearSVC vs Logistic Regression on 20 Newsgroups (Part B)

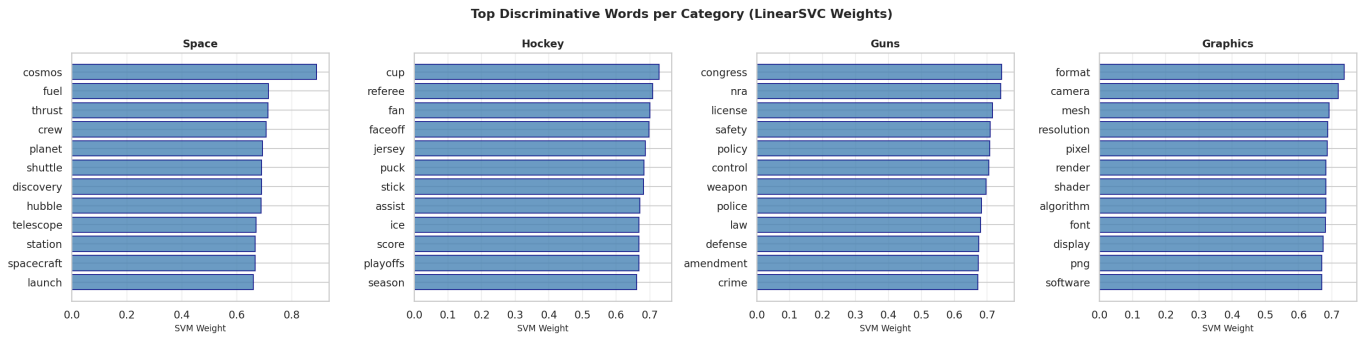

Figure 5: Top Discriminative Words per News Category — LinearSVC Weights (Part B)

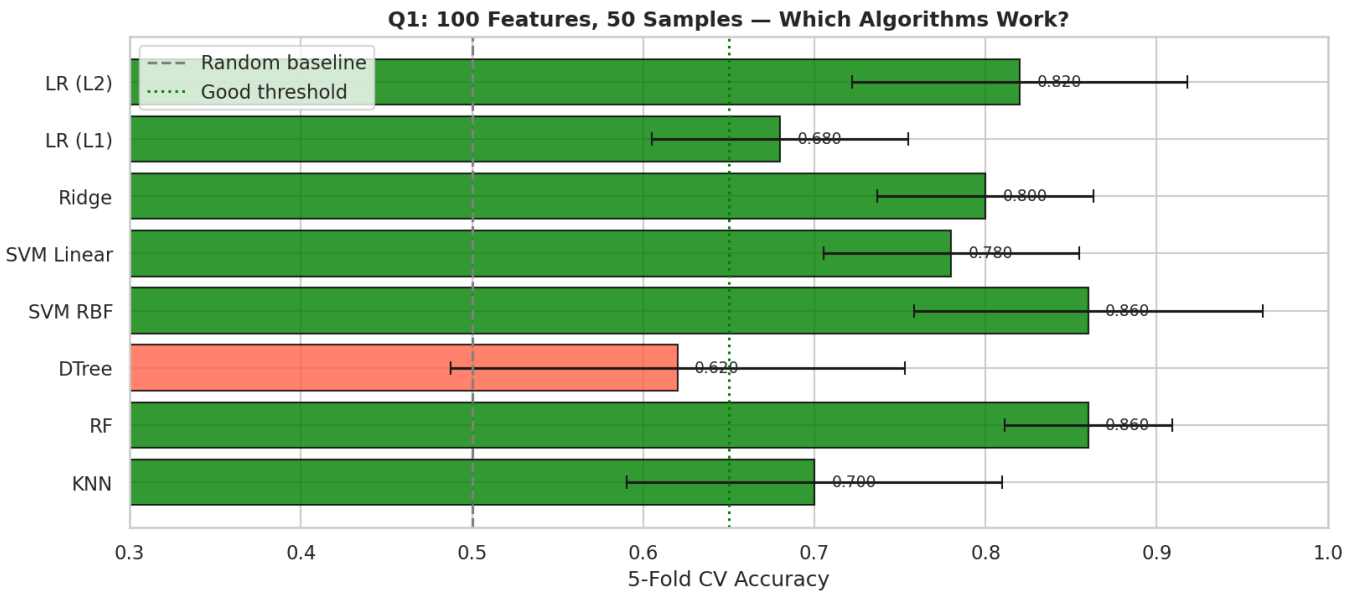

Figure 6: Q1 — Which Algorithms Survive 100 Features with 50 Samples? (Part C)

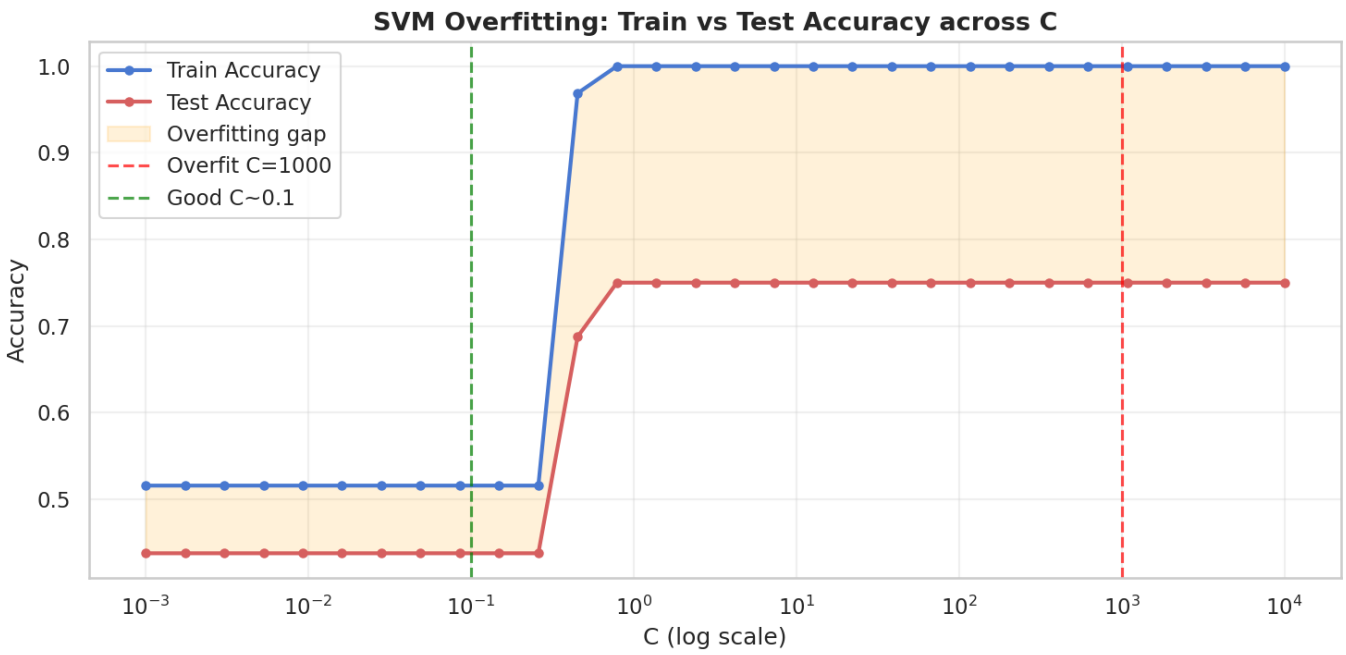

Figure 7: Q3 — SVM Overfitting: Train vs Test Accuracy across C Values (Part C)

Algorithm Selection Guide — Week 6 ML

Scenario	Best Choice	Why	Avoid
Text / NLP (TF-IDF)	LinearSVC / LR	Sparse high-dim space	RF, KNN
n < 200, d > 100	LR (L1), SVM(linear)	Regularisation prevents overfit	XGB, KNN
Tabular, max acc	XGBoost	SOTA on tabular data	KNN
Need interpretability	Logistic Regression	Coefficients = feature impact	RF, SVM
Missing values	XGBoost	Native handling	KNN, SVM
Non-linear boundary	RF or SVM (RBF)	Non-parametric / kernel	LR, Ridge
Need probabilities	LR or RF	Calibrated outputs	SVM, Ridge
Large n > 50K	LR, LinearSVC	Scales linearly	SVM(RBF), KNN
Fast robust baseline	Random Forest	Low variance, parallel	DT alone
No feature scaling	DT, RF, XGBoost	Tree splits are scale-free	KNN, SVM, LR

Figure 8: Algorithm Selection Guide — Scenario → Best Model → What to Avoid (Part D)